

SWNT Raman analysis

Single-wall carbon nanotube spectra, using the `ramanfit_swnt` package:

load → **Si calibration** → **RBM** (→ **diameter, candidate (n,m)**) → **joint D-G fit**
→ **2D** → **quality metrics**

Unlike the amorphous/MW-carbon D/G fit in `ramanfit.ipynb`, this:

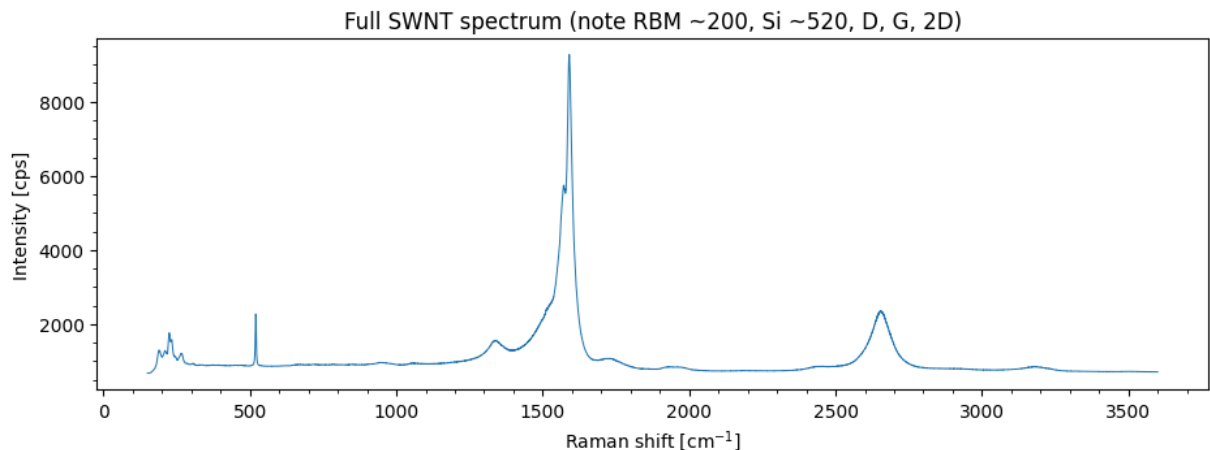
- models the low-frequency **RBM** band (diameter fingerprint),
- fits the whole **D-G region jointly over 1000–1700 cm^{-1}** with four Lorentzians — **D**, a broad **D'** disorder band (~ 1500 – 1535), and the **G**–**G**⁺ **doublet** — so the D-to-G valley is modelled instead of forced onto a straight baseline ($R^2 \approx 0.998$ vs ≈ 0.98 for isolated windows),
- calibrates the shift axis against the **Si 520.7 cm^{-1}** substrate line.

```
In [1]: import matplotlib.pyplot as plt
from ramanfit_swnt import load_spectrum, analyze_swnt, SwntConfig
from ramanfit_swnt.report import plot_analysis, summary_text

INFILE = "data/LA2026062501.tsv" # 532 nm SWNT spectrum on Si
x, y = load_spectrum(INFILE)
print(f"{len(x)} points, {x[0]:.0f}–{x[-1]:.0f}  $\text{cm}^{-1}$ ")
```

25732 points, 150–3600 cm^{-1}

```
In [2]: # Raw spectrum overview
fig, ax = plt.subplots(figsize=(11, 3.5))
ax.plot(x, y, lw=0.7)
ax.set(xlabel="Raman shift [ $\text{cm}^{-1}$ ]", ylabel="Intensity [cps]",
       title="Full SWNT spectrum (note RBM ~200, Si ~520, D, G, 2D)")
ax.minorticks_on()
plt.show()
```



Configure and run the pipeline

Adjust `laser_nm` (532/633/785), `gminus_lineshape` (`lorentzian` for semiconducting, `bwf` for metallic tubes) or any region in `SwntConfig`.

```
In [3]: cfg = SwntConfig(laser_nm=532, gminus_lineshape="lorentzian")
res = analyze_swnt(x, y, cfg)
print(summary_text(res))
```

Si calibration: observed 519.76 cm-1, offset -0.94 cm-1 (nominal 520.7)

RBM peaks (diameter, candidate (n,m)):

191.2 cm-1	d=1.30 nm	-
210.2 cm-1	d=1.18 nm	(15,0)M, (14,2)M, (10,7)M
225.2 cm-1	d=1.10 nm	(11,5)M, (8,8)M, (12,3)M
233.6 cm-1	d=1.06 nm	(13,1)M, (12,3)M, (8,8)M
246.6 cm-1	d=1.01 nm	(9,6)M, (13,1)M
266.2 cm-1	d=0.93 nm	-
305.9 cm-1	d=0.81 nm	(9,2)S, (10,0)S, (8,3)S

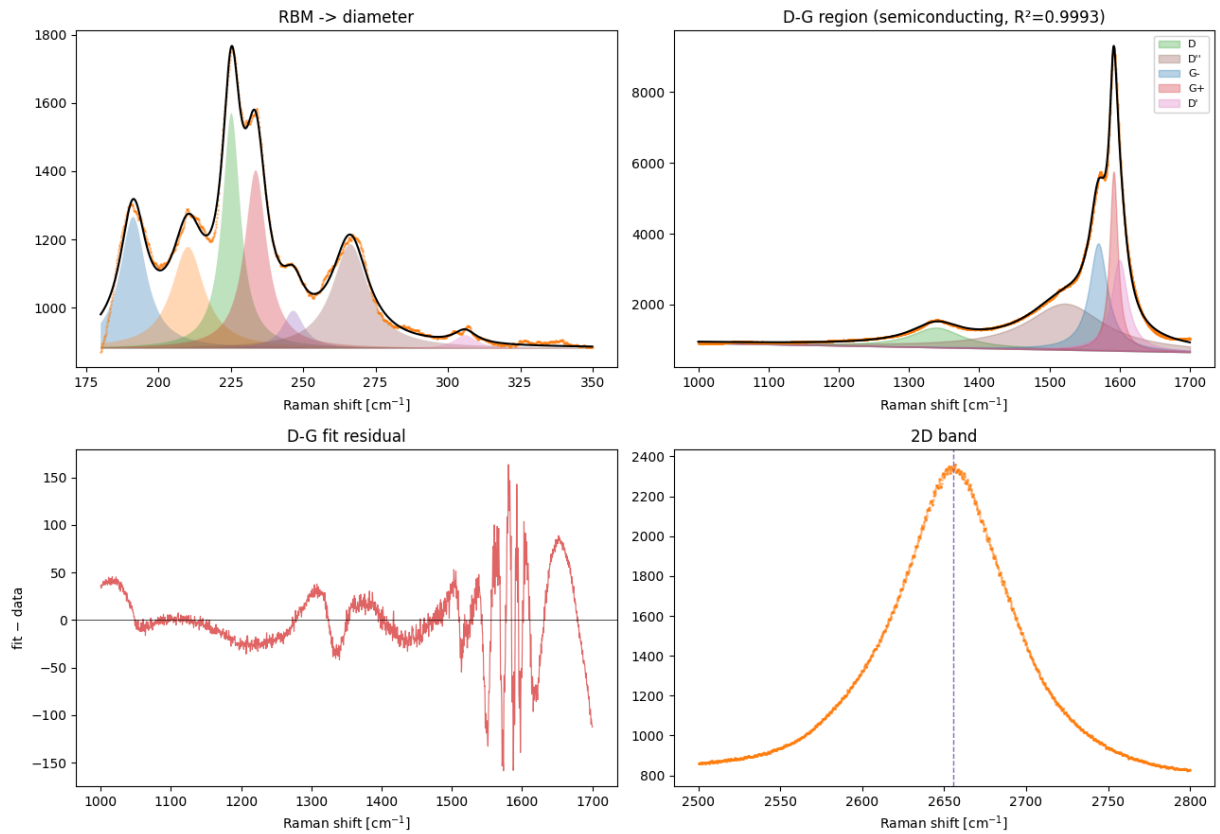
D-G joint fit ($R^2=0.9993$):

D 1338.9 cm-1 (FWHM 88)
D'' 1522.1 cm-1 (FWHM 132, broad disorder band)
G+ 1591.4, G- 1569.5 (split 21.9 cm-1) -> semiconducting
D' 1599.5 cm-1 (FWHM 27)

$I(D)/I(G+) = 0.110$

2D band: 2655.5 cm-1, $I(2D)/I(G+) = 0.320$

```
In [4]: # Four-panel decomposition: RBM / D-G joint fit / D-G residual / 2D
fig = plot_analysis(res)
plt.show()
```



RBM → diameter → candidate (n,m)

(n,m) candidates come from a first-order tight-binding Kataura model and are **approximate** — see `ramanfit_swnt/kataura.py`.

```
In [5]: for pk in res["rbm"]["peaks"]:
         nm = ", ".join(f"({c['n']},{c['m']}){c['type']}"
                        for c in pk.get("nm_candidates", [])[:3]) or "-"
         print(f"RBM {pk['center']:7.1f} cm⁻¹    d = {pk['diameter_nm']:.3f} nm
```

RBM	191.2 cm ⁻¹	d = 1.297 nm	—
RBM	210.2 cm ⁻¹	d = 1.180 nm	(15,0)M, (14,2)M, (10,7)M
RBM	225.2 cm ⁻¹	d = 1.101 nm	(11,5)M, (8,8)M, (12,3)M
RBM	233.6 cm ⁻¹	d = 1.062 nm	(13,1)M, (12,3)M, (8,8)M
RBM	246.6 cm ⁻¹	d = 1.006 nm	(9,6)M, (13,1)M
RBM	266.2 cm ⁻¹	d = 0.931 nm	—
RBM	305.9 cm ⁻¹	d = 0.811 nm	(9,2)S, (10,0)S, (8,3)S

D-G band peaks (joint fit)

Centers, widths and uncertainties for the four components fit jointly over 1000–1700 cm⁻¹. The broad **D'** band fills the D-to-G valley; finite `center ±` values confirm the fit is well-determined (not railed).

```
In [6]: dg = res["dg"]
         print(f"D-G joint fit R² = {dg['r_squared']:.4f} ({'metallic' if dg['metal
```

```

print(f"{'peak':4s} {'center':>9s} {'± (cm-1):>9s} {'FWHM':>7s} {'height':>7s} ")
for rec in (dg['D'], dg['Dpp'], dg['G_minus'], dg['G_plus'], dg['Dprime']):
    ce = rec.get('center_stderr')
    ce_s = f"{ce:.2f}" if ce is not None else " n/a"
    print(f"{'rec['label']':4s} {'rec['center']':9.1f} {'ce_s':>9s} "
          f"{'rec.get('fwhm', float('nan'))':7.1f} {'rec['height']':9.0f}")

```

D-G joint fit $R^2 = 0.9993$ (semiconducting)

peak	center	± (cm ⁻¹)	FWHM	height
D	1338.9	0.21	88.4	559
D''	1522.1	0.43	132.4	1304
G-	1569.5	0.04	30.9	3025
G+	1591.4	0.03	13.9	5067
D'	1599.5	0.24	27.4	2570

Si calibration & quality metrics

```

In [7]: cal = res["calibration"]
g = res["gband"]
q = res["quality"]
print(f"Si calibration applied: {cal['applied']} (offset {cal['offset']:+.2f} cm-1)")
print(f"G+ = {g['G_plus']['center']:.1f}, G- = {g['G_minus']['center']:.1f} cm-1")
print(f"character: {'metallic' if g['metallic'] else 'semiconducting'}")
print(f"I(D)/I(G+) = {q['ID_IG']:.3f} (low → high quality / few defects)")
print(f"2D = {q['twod_center']:.1f} cm-1, I(2D)/I(G+) = {q['I2D_IG']:.3f}")

```

Si calibration applied: True (offset -0.94 cm⁻¹)

G⁺ = 1591.4, G⁻ = 1569.5 cm⁻¹

character: semiconducting

I(D)/I(G⁺) = 0.110 (low → high quality / few defects)

2D = 2655.5 cm⁻¹, I(2D)/I(G⁺) = 0.320

Save a summary

```

In [8]: with open("swnt_summary.txt", "w") as f:
        f.write(summary_text(res))
        print("wrote swnt_summary.txt")

```

wrote swnt_summary.txt